Lemma. Let $M \subset \mathbb{R}^3$ be a surface, and let $\alpha: I \to M$ be a curve.
If for some patch $X: D \to \mathbb{R}^3$, range $\alpha \subseteq X[D]$,
then there exist unique differentiable functions α_1, α_2 on I such that

$$\alpha(t) = X(\alpha_1(t), \alpha_2(t)) \quad \text{for all } t \in I.$$

Proof. By definition